



18-Port Full Gigabit L2+ Managed Industrial Ethernet Switch (NAV-I-16R2S)

Full gigabit access with 2x SFP Fiber Uplinks

The NAV-I-16R2S is a full gigabit managed industrial Ethernet fiber switch designed for high-performance environments, supporting non-blocking wire-speed forwarding and L2+ network management.



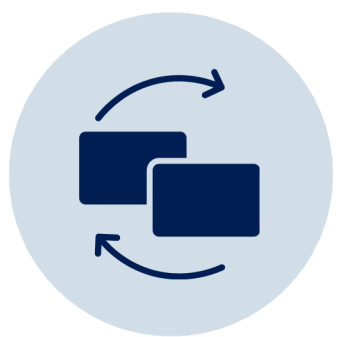
Gigabit Access

Features 16x 10/100/1000Base-T RJ45 ports and 2x 100/1000Base-X uplink SFP fiber ports to meet various networking scenarios.



Robust Management

Supports IPv4/IPv6 management, static route forwarding, and rich VLAN functions (Voice VLAN, QinQ) for easy maintenance.



Industrial Reliability

Supports ERPS ring network (<20ms recovery) and STP/RSTP/MSTP (<50ms) to eliminate loops and ensure uninterrupted communication.

Switching Capacity

52Gbps

Non-blocking

Forwarding Rate

26.78Mpps

@64byte

Engineered for harsh industrial applications, this switch features IP40 protection, a fanless galvanized steel metal shell, and a wide operating temperature range of -40 to +80°C. Ideally suited for intelligent transportation, rail, power industry, and green energy construction.

18-Port Gigabit L2+ Managed Industrial Ethernet Switch (NAV-I-16R2S)



Product Overview

The **NAV-I-16R2S** is a fully managed industrial Ethernet fiber switch designed for high-performance environments. It features 16x 10/100/1000Base-T adaptive RJ45 ports and 2x 100/1000Base-X uplink SFP fiber ports, with every port supporting wire-speed forwarding.

This device provides comprehensive L2+ network management capabilities, including IPv4/IPv6 management, static route forwarding, and advanced security mechanisms. It supports a wide range of ACL/QoS policies and rich VLAN functions for streamlined management. To ensure network reliability, it supports multiple redundancy protocols such as STP/RSTP/MSTP (<50ms) and ERPS (ITU-T G.8032) ring network protocol (<20ms), ensuring rapid recovery in the event of a link failure.

The NAV-I-16R2S is engineered for harsh industrial applications, including intelligent transportation, rail, power utilities, mining, and green energy construction.

Key Features

Gigabit Access & Uplink

- 16× Gigabit RJ45 ports + 2× SFP fiber uplinks for flexible networking.
- Supports non-blocking wire-speed forwarding and IEEE 802.3x full-duplex flow control.

Secure Network Control

- Features port isolation, storm control, and IP+MAC+Port+VLAN binding.
- Includes 802.1X authentication and flexible ACL policies for strict access management.

Intelligent Traffic Handling

- Supports ERPS ring (<20ms) and STP/RSTP/MSTP for rapid redundancy and loop elimination.
- L2+ functions include VLANs, QoS (SP/WRR), and IGMP Snooping for video optimization.

Industrial-Grade Design

- Fanless IP40 galvanized steel shell; operates reliably in -40 to +80°C environments.
- Dual redundant power inputs (AC priority + DC backup) ensure continuous uptime.

Versatile Maintenance

- Comprehensive management via Web (HTTPS), CLI, Telnet, SSH, and SNMP (V1/V2/V3).
- Includes RMON, system logs, and cable diagnostics for easy network optimization.

A robust L2+ managed switch featuring 16 Gigabit ports and 2 Fiber uplinks, engineered for harsh industrial sites. It combines high-speed forwarding with dual-redundant power, advanced security, and rapid self-healing ring technology for critical infrastructure.

Technical Specification

Feature	Description
Model	NAV-I-16R2S
Interface Characteristics	1Console port (115200,N,8,1) Power-off alarm switch (FAULT) 2AC100-240V power input ports
Fixed Port	16*10/100/1000Base-T RJ45 ports (Data) 2*100/1000Base-X uplink SFP fiber ports (Data) 2*DC12-48V power input ports (support anti-reverse protection)
Ethernet Port	Port 1-16 support 10/100/1000Base-T(X) auto-sensing, full/ half duplex MDI/MDI-X self-adaption
Twisted Pair Transmission	10BASE-T: Cat3,4,5 UTP (≤100 meters) 100BASE-TX: Cat5 or later UTP (≤100 meters) 1000BASE-T: Cat5e or later UTP (≤100 meters)
Optical Fiber Port	Gigabit optical fiber interface, default no include optical module (optional single-mode/multi-mode, single fiber/ dual fiber optical module. LC)
Optical Cable/Distance	Multi-mode: 850nm/0-500m Single-mode: 1310nm/0-40km, 1550nm/0-120km
Network Management Type	L2+
Network Protocol	IEEE 802.3 10BASE-T, IEEE 802.3i 10Base-T, IEEE 802.3u 100Base-TX, IEEE 802.3ab 1000Base-T, IEEE 802.3z 1000Base-X, IEEE 802.3x
Forwarding Mode	Store and Forward (Full Wire Speed)
Switching Capacity	52Gbps (non-blocking)
Forwarding Rate	26.78Mpps (@64byte)
MAC Address Table	8K
Buffer Memory	6M
Jumbo Frame	9.6K
LED Indicator	Power: PWR (Yellow) System: SYS (Yellow) Network: Link (Yellow) Rate: SPEED (Green) Fiber port: L/A (Green)
Reset Switch	Yes, Press and hold the reset switch for 10s and release it to restore the factory settings

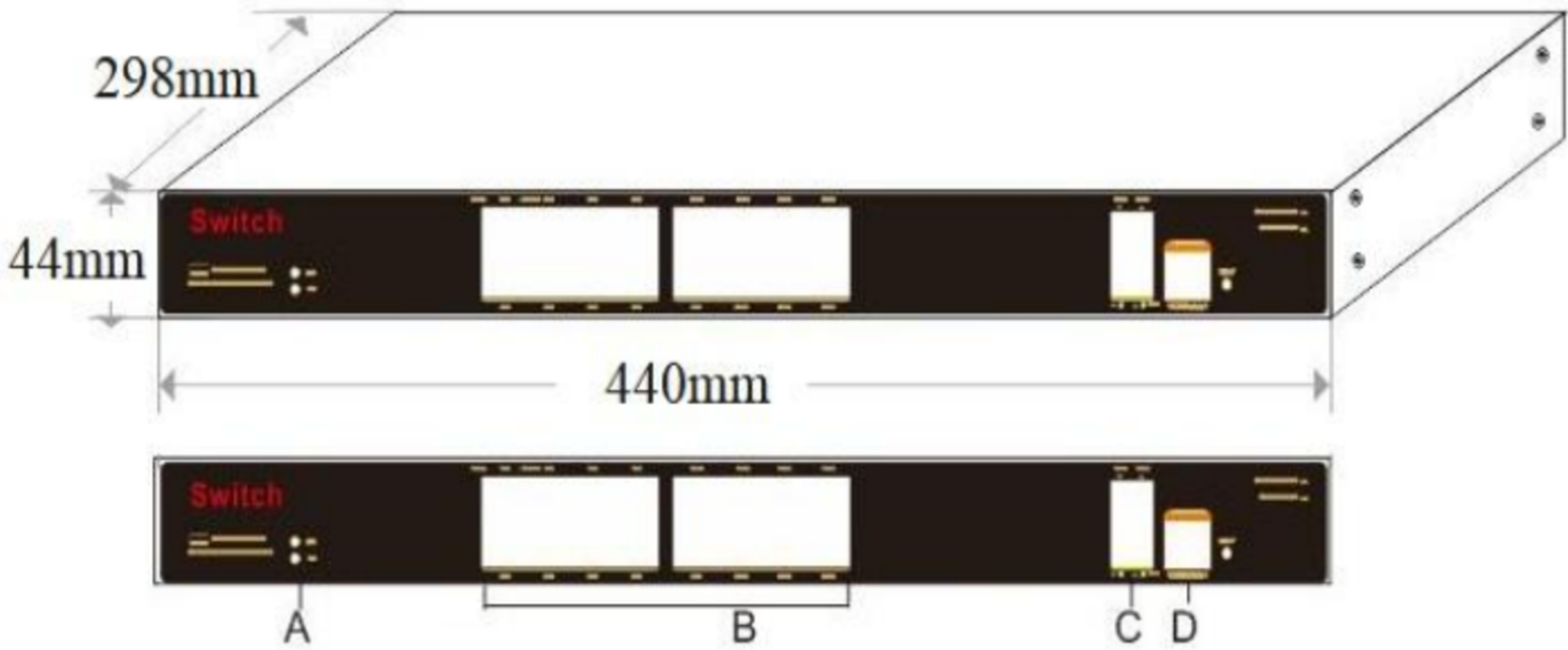
Power Supply	Total PWR / Input Voltage: 30W / (AC100-240V) Power Consumption: Standby<16W, Full load<30W Built-in Power: AC100~240V 50-60Hz, 0.56A Ports: Alarm switch port, 2 AC power input ports, 2*DC12-48V input ports Redundancy: Dual input power port design: AC power supply priority to support anti-reverse connection protection, and automatically switches to DC connection when power fails.
Physical Parameter	Operation TEMP/Humidity: -40~+80°C; 5%~90% RH Non condensing Storage TEMP/Humidity: -40~+85°C; 5%~95% RH Non condensing Dimension (L*W*H): 440*298*44mm Net/Gross Weight: 3.75kg/4.4kg Installation: Desktop, 1U/19" cabinet
Certification & Warranty	Protection Level: IP40 Lightning Protection: 6KV 8/20us IEC Standards: - IEC61000-4-3(RS): 10V/m (80~1000MHz) - IEC61000-6-2 (Common Industrial Standard) - IEC61000-4-9 (Pulsed magnet field): 1000A/m - IEC61000-4-10 (Damped oscillation): 30A/m, 1MHz - IEC61000-4-12/18 (Shockwave): CM 2.5kV, DM 1kV - IEC61000-4-4 (EFT): Power cable: ±4kV, Data cable: ±2kV - IEC61000-4-16 (Common-mode transmission): 30V, 300V, 1s - IEC61000-4-2 (ESD): ±8kV contact discharge, ±15kV air discharge - IEC61000-4-6 (Radio frequency transmission): 10V (150kHz~80MHz) - IEC61000-4-8 (Power frequency magnetic field): 100A/m, 1000A/m, 1-3s - IEC61000-4-5 (Surge): Power cable: CM±4kV/DM±2kV, Data cable: ±4kV FCC/CISPR: FCC Part 15/CISPR22(EN55022): Class B
Mechanical Properties	IEC60068-2-6 (Anti Vibration), IEC60068-2-32 (Free Fall), IEC60068-2-27 (Anti Shock)
Certification	CCC, CE mark, commercial, CE/LVD EN62368-1, FCC Part 15 Class B, RoHS
Warranty	5 years, lifelong maintenance.

Network Management Feature

Interface	Port temperature protection settings IEEE 802.3x flow control (Full duplex) Broadcast storm control based on port speed Port EEE Green Ethernet energy-saving configuration Limit the speed of the message traffic at the inbound and outbound ports, with a mini granularity of 64Kbps
L3 Features	IPv4/IPv6 static route/ default route max 128 entries ARP protocol, max 1024 entries, L2+ network management function L3 routing and forwarding, support communication between different network segments and different VLAN
VLAN	Port configuration of Access, Trunk, Hybrid VLAN based on MAC, VLAN based on the protocol (4K) VLAN based on port, IEEE802.1q. Voice VLAN, QinQ configuration
Port Aggregation	Static aggregation, LACP dynamic aggregation Max 9 aggregation groups, per group support up to 8 ports
Spanning Tree	STP (IEEE802.1d), RSTP (IEEE802.1w), MSTP (IEEE802.1s)
Industrial Ring Network Protocol	Support ERPS ring network, Recovery time less than 20ms, 255 Ring at most, max 254 devices per ring
Multicast	IGMP Snooping v1/v2/v3, Max 1024 multicast groups MLD Snooping, Multicast VLAN, User quick exit mechanism
Port Mirroring	Bidirectional data mirroring based on port
QoS	Queue Scheduling Algorithm (SP, WRR, SP+WRR) Flow-based rate limiting, Flow-based packet filtering Flow-based redirection, Diff-Serv QoS, Priority Mark/ Remark 8*Output queues of each port, 802.1p/DSCP priority mapping
ACL	ACL issued based on port and VLAN L2-L4 packet filtering function can match the first 80 bytes of the message and provide ACL definition based on source MAC address, destination MAC address, source IP address, destination IP address, IP protocol type, TCP/UDP port, TCP/UDP port range, VLAN, etc.

Security	MAC address the black hole Port broadcast message suppression Port-based IEEE802.1X authentication MAC address the learning number limit SSL ensures data transmission security IP+MAC+VLAN+port four-way binding function Port isolation, ARP message rate limit function DoS attack prevention, ARP intrusion detection function User hierarchical management and password protection AAA & RADIUS & TACACS+ certification, ARP inspection IP source address protection, Host data backup mechanism SSH 2.0 provides a secure encrypted channel for user login
DHCP	DHCP Client, DHCP Snooping, DHCP Server, DHCP Relay
Management	System work log, Link Layer Discovery Protocol NTP clock, Cable length detection, SNMP V1/V2/V3 Ping detection, Web network management (HTTPS) NMS platform cluster management (LLDP+SNMP) One click recovery, View CPU real-time utilization status FTP, TFTP, Xmodem, SFTP file upload and download management Console/ AUX Modem/ Telnet/ SSH2.0 CLI command line configuration
System	Web browser: Mozilla Firefox 2.5 or higher, Google Chrome V42 or higher, Cat5 and above Ethernet cable TCP/IP, network adapter, and network operating system (such as Microsoft Windows, Linux, Mac OS X) installed on each computer in the network

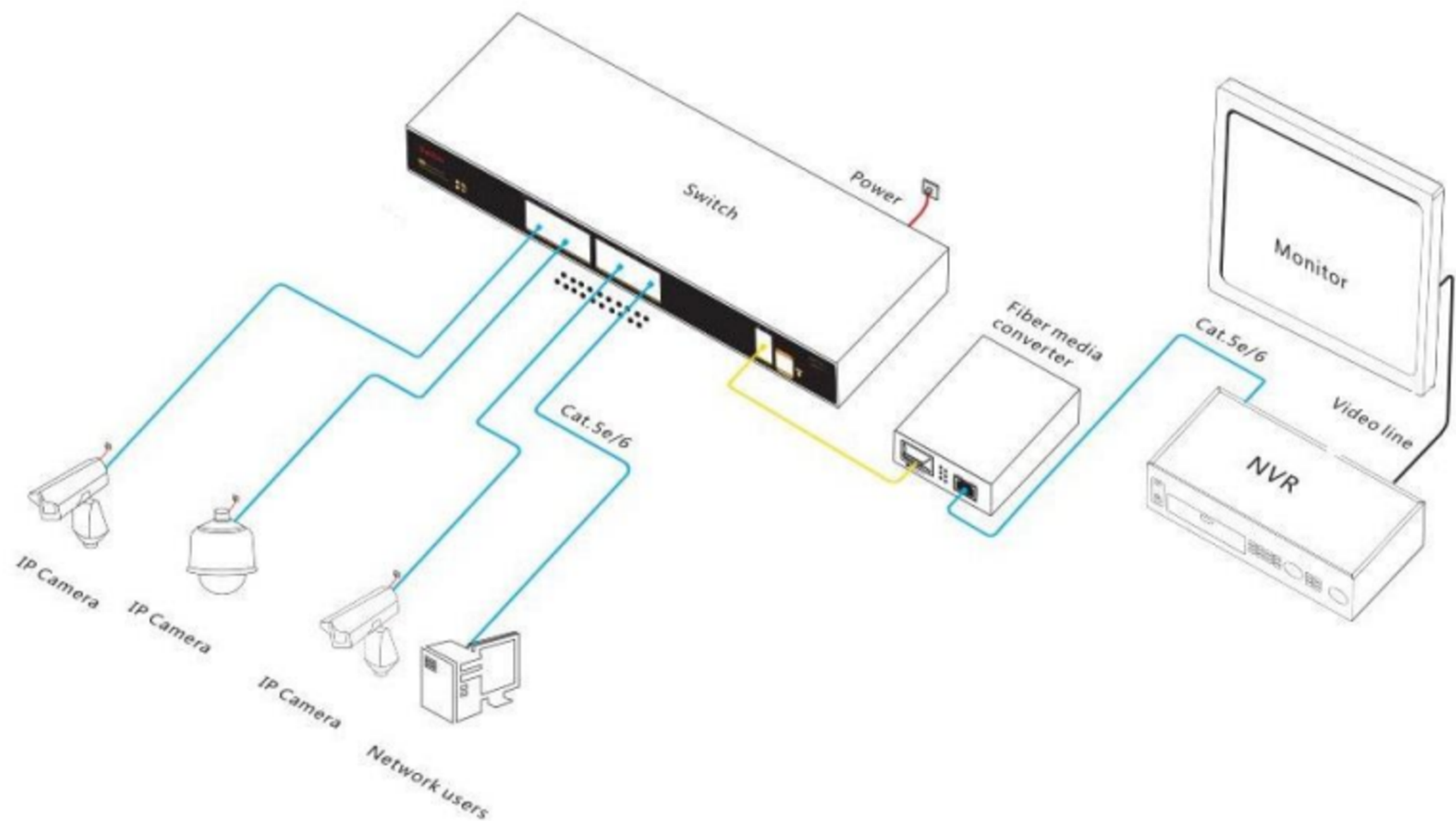
Dimensions



- A. Working indicator
- B. 16*10/100/1000M RJ45 ports

- C. 2*100/1000M uplink SFP fiber ports
- D. Console port

Applications



Ordering Information

Model	Description	Built-in Power Supply
NAV-I-16R2S	L2+ managed industrial Ethernet switch with 16*10/100/1000M RJ45 ports and 2*100/1000M uplink SFP fiber ports. It has a built-in power supply and supports 2AC+DC power input and 1U/19-inch cabinet installation.	2*30W
Note: The optical module is not included and needs to be purchased separately.		

Packing List

Content	Qty	Unit
18-port full gigabit L2+ managed industrial Ethernet switch	1	SET
AC Power Cable	1	PC
RJ45-DB9 Line	1	PC
Mounting Kits	1	SET
User Guide	1	PC
Warranty Card	1	PC

Optical Module

Product	Model	Description	Unit
1.25G Optical Module	2630-G	Industrial SFP optical module, 1.25G multi-mode dual fiber 850nm, transmission distance: 550m, LC interface. supports DDM function and hot plugging.	PC
	2632-G	Industrial SFP optical module, 1.25G single-mode dual fiber 1310nm, transmission distance: 20km, LC interface. supports DDM function and hot plugging.	PC
	2612-T-G	Industrial SFP optical module, 1.25G single-mode single fiber TX1310nm/RX1550nm, transmission distance: 20km, LC interface. supports DDM function and hot plugging.	PC
	2613-RG-	Industrial SFP optical module, 1.25G single-mode single fiber TX1550nm/RX1310nm, transmission distance: 20km, LC interface, supports DDM function and hot plugging.	PC
	2612-T-SCG-	Industrial SFP optical module, 1.25G single-mode single fiber TX1310nm/RX1550nm, transmission distance: 20km, SC interface. supports DDM function and hot plugging.	PC
	2613-R-SC-G	Industrial SFP optical module, 1.25G single-mode single fiber TX1550nm/RX1310nm, transmission distance: 20km, SC interface. supports DDM function and hot plugging.	PC
Power Module	2633	1.25G SFP optical module transfers to 10/100/1000M RJ45 port.	PC